

Statistics II: Introduction to Inference

Module 5: Large Sample Theory

1 Sampling distributions

Recall the sampling distributions of some important statistics.

Example 1. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \mathcal{N}(\mu, \sigma^2)$. Consider the sample variance S_n^2 (Sample variance), and the sampling distribution of sample variance, which is $\sigma^2 \chi_{n-1}^2 / n$. The sampling distribution of S_n^2 for $\mu = 0, \sigma^2 = 10$ is given in Figure 1.

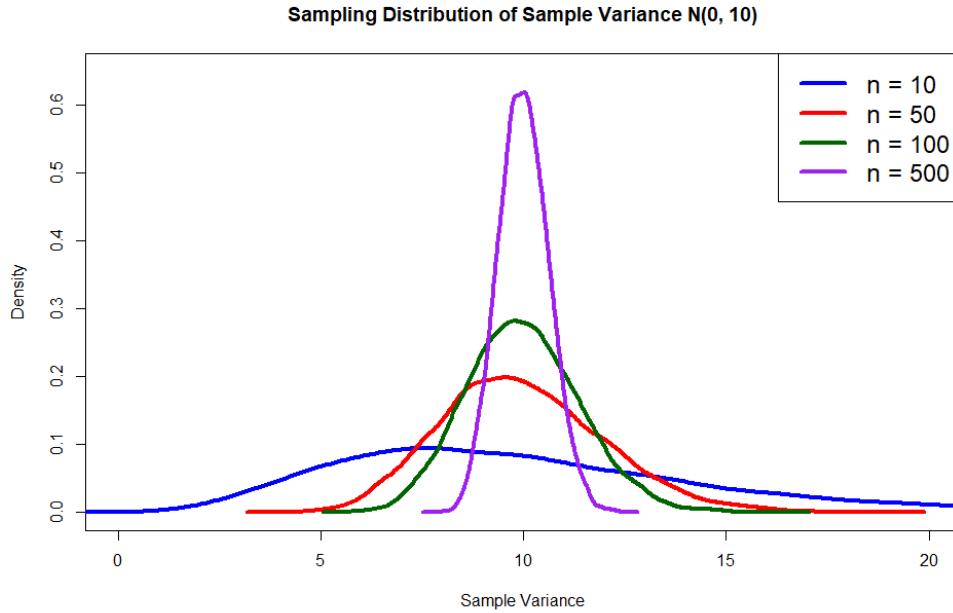


Fig. 1: Sampling distribution of sample variance from `normal(0, 10)` distribution.

Example 2. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$. Consider the sample mean $\bar{X}_n$ and its sampling distribution, which is $n^{-1} \text{Gamma}(n\alpha, \beta)$. The sampling distribution of $\bar{X}_n$ for $\alpha = 10, \beta = 10$ is given in Figure 2.

Example 3. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{uniform}(0, \theta)$. Consider the maximum order statistics $T_n = X_{(n)}$. The sampling distribution of T_n for $\theta = 25$ is given in Figure 3.

Remark 1. In the above examples the following can be observed.

- (a) The distribution of good statistics or estimators changes as n changes.
- (b) Usually, a good statistic or estimator, say, T_n is a function of $X_1, \dots, X_n$. For example,

$$T_{11} = X_1, \quad T_{12} = (X_1 + X_2)/2, \quad T_{13} = (X_1 + X_2 + X_3)/3, \quad \text{and so on} \quad T_n = \bar{X}_n.$$

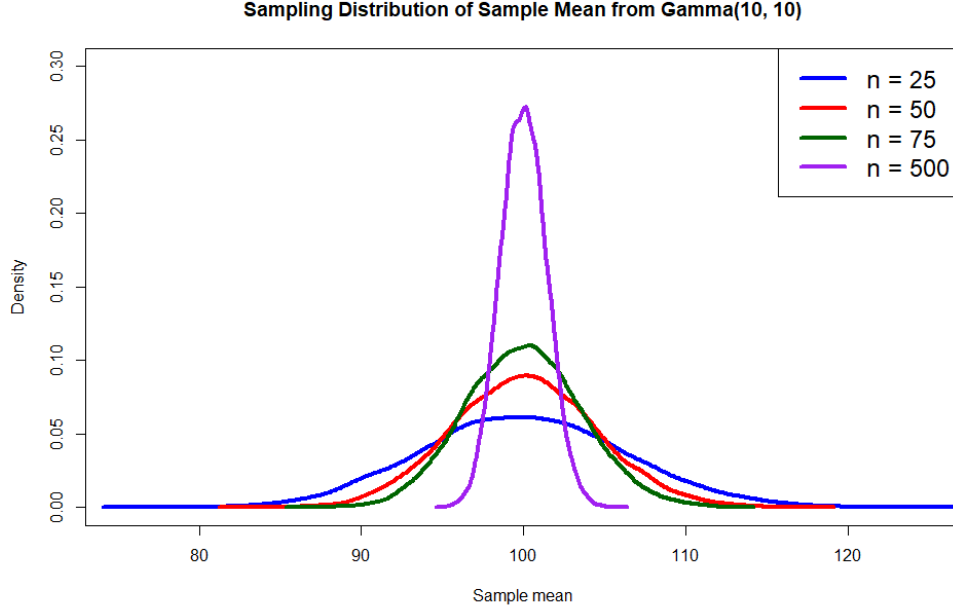


Fig. 2: Sampling distribution of sample mean from $\text{Gamma}(10, 10)$ distribution.

(c) Therefore, it is natural to ask, *how the statistic behaves when we have a large number of representative samples?*

(d) In this module, we will learn the *large sample behavior of statistic*.

2 Convergence of a sequence of random variables

Towards understanding the large sample behavior of a statistic, say T_n , it is important to understand what do we mean by convergence of a sequence of random variable.

Example 4. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$ and consider the statistic $T_n = \bar{X}_n$. The distribution of T_n changes over n as follows:

$$T_1 = \begin{cases} 0 & \text{w.p. } (1-p) \\ 1 & \text{w.p. } p \end{cases}, T_2 = \begin{cases} 0 & \text{w.p. } (1-p)^2 \\ \frac{1}{2} & \text{w.p. } 2p(1-p) \\ 1 & \text{w.p. } p^2 \end{cases}, T_3 = \begin{cases} 0 & \text{w.p. } (1-p)^3 \\ 1/3 & \text{w.p. } 3p(1-p)^2 \\ 2/3 & \text{w.p. } 3p^2(1-p) \\ 1 & \text{w.p. } p^3 \end{cases} \quad \text{and so on.}$$

So, one can not directly generalize the concept of convergence of real sequence for convergence of random variables (RV). [Recall, a real sequence $\{a_n\}$ converges to a point a if for any $\epsilon > 0, \exists n_\epsilon$ s.t. $|a_n - a| < \epsilon$ for $n \geq n_\epsilon$.] Therefore, to define convergence of a sequence of RVs to a RV, i.e., to define $T_n \rightarrow T$, where $\{T_n\}$ and T are RVs, one first,

- reduces the sequence of random variable indicating difference of T_n and T , $|T_n - T|$, to a sequence of real numbers,
- then apply the concept of real convergence.

2.1 Convergence in r -th mean.

A natural way to reduce the sequence of random variables $|T_n - T|$ to a sequence of real numbers, $\xi_n = E(|T_n - T|)$, and say that $T_n \rightarrow T$ as $n \rightarrow \infty$, if $\xi_n \rightarrow 0$ as $n \rightarrow \infty$. This leads to the idea of

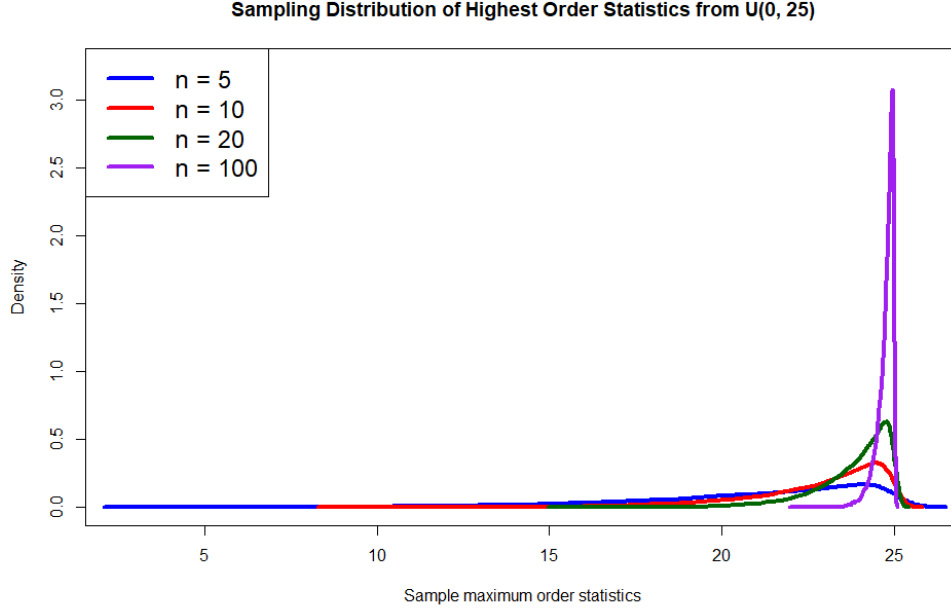


Fig. 3: Sampling distribution of maximum order statistics of $\text{uniform}(0, 25)$ distribution.

convergence in r -th mean.

Definition 1 (Convergence in r -th mean). Let T_n be a sequence of random variables, and T be a random variable such that $E(|T_n|^r) < \infty$ for all n , and $E(|T|^r) < \infty$ for some $r \geq 1$. Then $T_n \rightarrow T$ in r -th mean if $E(|T_n - T|^r) \rightarrow 0$ as $n \rightarrow \infty$. Notationally, $T_n \xrightarrow{L^r} T$ as $n \rightarrow \infty$.

Remark 2. Sometimes the limiting RV, T , is of degenerate type, i.e., $P(T = c) = 1$. In such cases, $T_n \xrightarrow{L^r} T$ can equivalently be expressed as $T_n \xrightarrow{L^r} c$.

Example 5. Let $X_1, \dots, X_n$ be a random sample from some random variable, satisfying $E(X_1) = \mu$ and $\text{var}(X_1) = \sigma^2 < \infty$. Then $\bar{X}_n \xrightarrow{L^2} \mu$.

Proof. To see this, observe that $E(\bar{X}_n) = \mu$, and hence

$$E(|\bar{X}_n - \mu|^2) = \text{var}(\bar{X}_n) = \frac{\sigma^2}{n} \rightarrow 0, \text{ as } n \rightarrow \infty.$$

Hence the result follows. $\square$

Remark 3. If $T_n \xrightarrow{L^r} T$ as $n \rightarrow \infty$ for some $r > 1$. Then $T_n \xrightarrow{L^s} T$ as $n \rightarrow \infty$ for all $1 \leq s \leq r$, i.e., convergence in higher order mean implies convergence in lower order mean.

The converse of the result is not true, i.e., convergence in lower order mean does not imply convergence in higher order mean.

Example 6. Let T_n , $n \geq 1$, be a sequence of random variables such that

$$T_n = \begin{cases} 0 & \text{with probability } 1 - 1/(n\sqrt{n}), \\ i & \text{with probability } 1/(n^2\sqrt{n}), \text{ for } i = 1, \dots, n. \end{cases}$$

Observe that,

$$E(|T_n|) = E(T_n) = \frac{1}{n^2\sqrt{n}} \{1 + \dots + n\} = \frac{n+1}{2n\sqrt{n}} \rightarrow 0.$$

Thus, $T_n \xrightarrow{L^1} 0$.

$$E(T_n^2) = \frac{1}{n^2\sqrt{n}}\{1^2 + \dots + n^2\} = \frac{(n+1)(2n+1)}{6n\sqrt{n}} \rightarrow \infty.$$

Thus, $T_n \not\rightarrow 0$ in L^2 .

Remark 4 (Continuous Mapping Theorem Does NOT Hold for L^r Convergence). *The above example also demonstrates that if $T_n \xrightarrow{L^r} T$, then it is NOT usually true that $g(T_n) \xrightarrow{L^r} g(T)$ where g is a continuous function. In this example, $g(x) = x^2$ is applied on T_n , and $T_n \xrightarrow{L^1} 0$, but $T_n^2 \not\xrightarrow{L^1} 0$ (i.e., $E(T_n^2) \not\rightarrow 0$ as $n \rightarrow \infty$).*

2.2 Convergence in probability.

Definition 2 (Convergence in probability). *Let $\{T_n\}$ be a sequence of RVs and T be another RV such that all of them are defined on the same probability space (so that the RV $T_n - T$ is well-defined), then we say that T_n converges in probability to T , notationally, $T_n \xrightarrow{P} T$ if for any $\epsilon > 0$,*

$$p_{n,\epsilon} = P(|T_n - T| > \epsilon) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

Remark 5. *Observe that, here $\{p_{n,\epsilon}\}$ is a real sequence measuring the difference between T_n and T (at ϵ level).*

Remark 6. *Sometimes the limiting RV, T , is of degenerate type, i.e., $P(T = c) = 1$. In such cases, $T_n \xrightarrow{P} T$ can equivalently be expressed as $T_n \xrightarrow{P} c$.*

Example 7. *Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$, and $T_n = X_{(n)}$. Show that $T_n \xrightarrow{P} \theta$.*

Fix any $\epsilon > 0$,

$$\begin{aligned} P(|T_n - \theta| < \epsilon) &= P(\theta - \epsilon < T_n < \theta + \epsilon) \\ &= \begin{cases} P(\theta - \epsilon < T_n \leq \theta) & \text{if } \epsilon \leq \theta \\ P(0 \leq T_n \leq \theta) & \text{if } \epsilon > \theta \end{cases} \\ &= \begin{cases} 1 - (\epsilon/\theta)^n & \text{if } \epsilon \leq \theta \\ 1 & \text{if } \epsilon > \theta \end{cases} \\ &\rightarrow 1 \text{ as } n \rightarrow \infty. \end{aligned}$$

Thus, $P(|T_n - \theta| > \epsilon) = p_{n,\epsilon} \rightarrow 0$ as $n \rightarrow \infty$.

Definition 3 (Consistency). *Let $\{T_n\}$ be a sequence of estimators for the parameter $\psi(\theta)$. Then $\{T_n\}$ is called a consistent sequence of estimators for $\psi(\theta)$ if $T_n \xrightarrow{P} \psi(\theta)$ as $n \rightarrow \infty$.*

Example 8. *In the above **Example 4**, $X_{(n)}$ is consistent for θ .*

Sufficient conditions for consistency: A sequence of estimators $\{T_n\}$ is consistent for $\psi(\theta)$, i.e., $T_n \xrightarrow{P} \psi(\theta)$ if

$$\mathbb{E}(T_n) \rightarrow \psi(\theta) \text{ as } n \rightarrow \infty \quad \text{and} \quad (1)$$

$$\text{Var}(T_n) \rightarrow 0 \text{ as } n \rightarrow \infty \quad (2)$$

Note: If (1) and (2) holds then $T_n \xrightarrow{P} \psi(\theta)$. But the converse is **NOT** true.

Proof. Observe that, for any $\epsilon > 0$,

$$\begin{aligned} P(|T_n - \psi(\theta)| > \epsilon) &= P(\{T_n - \psi(\theta)\}^2 > \epsilon^2) \\ &\leq \frac{\mathbb{E}[\{T_n - \psi(\theta)\}^2]}{\epsilon^2} \end{aligned}$$

Now,

$$\mathbb{E}[\{T_n - \psi(\theta)\}^2] \leq 3 \text{Var}(T_n) + 3[\mathbb{E}(T_n) - \psi(\theta)]^2 \rightarrow 0 \quad \text{by (1) and (2).}$$

□

Example 9 (WLLN: IID case with finite variance). Let $X_1, \dots, X_n$ be IID samples from some distribution with expectation μ and finite variance σ^2 . Then $\mathbb{E}(\bar{X}_n) = \mu$ and $\text{Var}(\bar{X}_n) = \sigma^2/n \rightarrow 0$ as $n \rightarrow \infty$. So, $\bar{X}_n \xrightarrow{p} \mu$ as $n \rightarrow \infty$.

2.3 Weak Law of Large Numbers (WLLN):

As shown in the last example, sample mean, $\bar{X}_n$, is a consistent estimator of population mean (expectation). This result is called law of large numbers (LLN). WLLN can be extended to the case with independent (not identically distributed) samples.

Result 1 (WLLN : Independent/Uncorrelated case). Let $X_1, \dots, X_n$ be independent samples such that $\mathbb{E}(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2; i = 1, \dots, n$. Further, let $\sigma_i^2 \leq M$ for some $M > 0$ for all $i = 1, \dots, n$. Then

$$\bar{X}_n - \bar{\mu}_n \xrightarrow{p} 0 \text{ as } n \rightarrow \infty, \quad \text{where } \bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mu_i.$$

Proof. Fix $\epsilon > 0$,

$$\begin{aligned} P(|\bar{X}_n - \bar{\mu}_n| > \epsilon) &\leq \frac{\mathbb{E}[(\bar{X}_n - \bar{\mu}_n)^2]}{\epsilon^2} \\ &= \frac{\mathbb{E}[\sum_{i=1}^n (X_i - \mu_i)^2 / n^2]}{\epsilon^2} \\ &= \frac{\sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j)}{\epsilon^2 n^2} \\ &= \frac{\sum_{i=1}^n \sigma_i^2}{n^2 \epsilon^2} \quad [\text{due to independence}] \\ &\leq \frac{nM}{n^2 \epsilon^2} \rightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

□

Remark 7. Observe that, in the above proof it is enough to have

$$\frac{\sum_{i=1}^n \sigma_i^2}{n^2} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Example 10. Let $X_i \sim \text{Bern}(p_i)$ for $i = 1, \dots, n$ and $X_1, \dots, X_n$ are mutually independent. Observe that,

$$\mathbb{E}(X_i) = p_i, \text{ and } \text{Var}(X_i) = p_i(1 - p_i) \leq \frac{1}{4} \quad \forall i.$$

Thus,

$$(\bar{X}_n - \bar{p}_n) \xrightarrow{p} 0 \text{ as } n \rightarrow \infty.$$

3 Convergence in distribution

Another concept of convergence of RVs is convergence in distribution.

Definition 4 (Convergence in distribution). Let $\{T_n\}$ be a sequence of RVs with CDFs $\{F_n\}$ and T be another random variable with CDF F . Then T_n is said to converge in distribution to T , notationally, $T_n \xrightarrow{d} T$ if

$$F_n(x) \longrightarrow F(x) \text{ as } n \rightarrow \infty,$$

for any continuity point x of $F(\cdot)$.

Remark 8. A point x is a continuity point of a CDF F if

$$F(x^-) = F(x) = F(x^+)$$

We know that any CDF F is right continuous (i.e., $F(x^+) = F(x)$ for all $x \in \mathbb{R}$), and non-decreasing. However, for some points $x \in \mathbb{R}$, it might happen that

$$F(x^-) < F(x) = F(x^+).$$

Such points are called discontinuity points of F .

Note: There can be at most countable number of discontinuity points in any CDF F .

Remark 9. If the limiting RV T is degenerate with $P(T = c) = 1$, then $T_n \xrightarrow{d} T$ can equivalently be written as $T_n \xrightarrow{d} c$.

Example 11. Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{uniform}(0, \theta)$ and $T_n = X_{(n)}$. Show that $T_n \xrightarrow{d} \theta$.

Proof. Let T be such that $P(T = \theta) = 1$. So, we'll show $T_n \xrightarrow{d} T$. The CDF of T , denoted by F_T , is as follows:

$$F_T(t) = \begin{cases} 0 & \text{if } -\infty < t < \theta \\ 1 & \text{if } t \geq \theta \end{cases}$$

Thus, F_T is discontinuous at the point θ only. So, the set of continuity points of F_T ,

$$\mathcal{C} = \{x : x \in \mathbb{R} \setminus \{\theta\}\}$$

Now, The CDF of T_n, F_n is given by

$$F_n(t) = \begin{cases} 0 & \text{if } -\infty < t < 0 \\ (t/\theta)^n & \text{if } 0 \leq t < \theta \\ 1 & \text{if } t \geq \theta \end{cases}$$

Consider any point $x \in \mathcal{C}$.

$$\begin{aligned} F_n(x) &= 0 = F_T(x) & \text{if } -\infty < x < 0, \\ F_n(x) &= (x/\theta)^n \rightarrow 0 = F_T(x) & \text{if } 0 \leq x < \theta, \\ F_n(x) &= 1 = F_T(x) & \text{if } x > \theta \end{aligned}$$

So, $T_n \xrightarrow{d} T$. □

3.1 Some important remarks

Remark 10. If $T_n \xrightarrow{p} T$ then $T_n \xrightarrow{d} T$, but the converse is not true in general. However, if $T_n \xrightarrow{d} c$ then $T_n \xrightarrow{p} c$.

Remark 11 (Continuous mapping theorem). If $T_n \xrightarrow{d/p} T$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function then $g(T_n) \xrightarrow{d/p} g(T)$.

Remark 12 (Slutsky's Lemma). Let $T_n \xrightarrow{d} T$ and $W_n \xrightarrow{p} c$ then

- (i) $T_n \pm W_n \xrightarrow{d} T \pm c$,
- (ii) $T_n W_n \xrightarrow{d} cT$, and
- (iii) if $c \neq 0$ then $T_n/W_n \xrightarrow{d} T/c$.

4 Central Limit Theorem (CLT)

The CLT concerns the large sample behavior of sample mean, after proper scaling (magnification). Like WLLN, there are various versions of CLT. Here we provide only the result for **IID** samples.

Theorem 1 (CLT, IID case). Let $X_1, \dots, X_n$ be IID samples from some distribution with expectation μ , and finite variance $\sigma^2 \in (0, \infty)$. Then,

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} = \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \xrightarrow{d} Z,$$

where $Z \sim N(0, 1)$ and $S_n = (X_1 + \dots + X_n)$.

Remark 13 (Asymptotic Distribution). Let T_n , $n \geq 1$, be a sequence of random variables. Then by the asymptotic distribution of T_n , we mean the sequences of real numbers $\{a_n\}$ and $\{b_n\}$, and a **non-degenerate** probability distribution F such that

$$\frac{T_n - a_n}{b_n} \xrightarrow{d} Z, \quad \text{where } Z \sim F.$$

The CLT provides the asymptotic distribution of $\bar{X}_n$, where $a_n = \mu$ and $b_n = \sigma/\sqrt{n}$, $n \geq 1$, and the non-degenerate probability distribution $N(0, 1)$.

Example 12 (Asymptotic distribution other than CLT). Let $X_1, \dots, X_n$ be IID samples from **uniform**(0, θ) distribution, and $T_n = X_{(n)}$. As $T_n \xrightarrow{p} \theta$, to find the asymptotic distribution of T_n , we center it around θ , and magnify the difference $\theta - T_n$ by n^r for some $r > 0$. Define $W_{n,r} = n^r(\theta - T_n)$, and $G_{n,r}$ as the cdf of $W_{n,r}$.

Observe that, for any $w > 0$ and sufficiently large n (such that $w < n\theta$),

$$G_{n,r}(w) = P(W_{n,r} \leq w) = P\left(T_n \geq \theta - \frac{w}{n^r}\right) = 1 - \left(1 - \frac{w}{\theta n^r}\right)^n = \begin{cases} 1 & \text{if } r < 1, \\ 1 - \exp\{-w/\theta\} & \text{if } r = 1, \\ 0 & \text{if } r > 1. \end{cases}$$

Thus the asymptotic distribution of T_n is given by $a_n = \theta$, $b_n = -1/n$ and the non-degenerate random variable **exponential**(1/ θ).

Example 13. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2)$ distribution. Then, $n^{-1} \sum_{i=1}^n X_i^2$ is an estimator of σ^2 . By CLT,

$$\frac{\sqrt{n} [\sum_{i=1}^n X_i^2 - \sigma^2]}{\sqrt{2\sigma^4}} \xrightarrow{d} Z \sim N(0, 1)$$

We say that $n^{-1} \sum_{i=1}^n X_i^2$ is asymptotically normally distributed with *asymptotic mean* σ^2 and *asymptotic variance* $2\sigma^4/n$.

Reference. Chapter 6 of *Probability and Statistics* by M. H. DeGroot and M. J. Schervish.